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Exp-14: Implement House price Prediction using appropriate machine learning algorithm

AIM

To implement a House Price Prediction Model using Linear Regression and predict the price of a house based on its features such as area, number of bedrooms, bathrooms, and number of floors.

ALGORITHM

1. Import the required Python libraries.
2. Load the house price dataset.
3. Display and examine the dataset.
4. Check for missing values and handle them.
5. Select the important features such as Area, Bedrooms, Bathrooms, and Floors.
6. Separate the input features X and target variable Y (House Price).
7. Split the dataset into training and testing data.
8. Create a Linear Regression model.
9. Train the model using the training data.
10. Predict house prices using the testing data.
11. Evaluate the model using Mean Squared Error (MSE) and R^2 score.
12. Give the predicted price for a new house.

PROGRAM

```
# Import required libraries

from sklearn.datasets import fetch_california_housing

from sklearn.model_selection import train_test_split

from sklearn.linear_model import LinearRegression

from sklearn.metrics import mean_squared_error, r2_score

# Load the California Housing dataset

housing = fetch_california_housing()

X = housing.data

y = housing.target

# Split the dataset into training and testing sets

X_train, X_test, y_train, y_test = train_test_split(

    X, y, test_size=0.2, random_state=42

)

# Create the Linear Regression model

model = LinearRegression()

# Train the model
```

```
model.fit(X_train, y_train)

# Predict house prices

y_pred = model.predict(X_test)

# Display results

print("Actual House Prices:")

print(y_test[:10])

print("\nPredicted House Prices:")

print(y_pred[:10])

# Evaluate the model

print("\nMean Squared Error:", mean_squared_error(y_test, y_pred))
print("R2 Score:", r2_score(y_test, y_pred))
```

Output:

```
...  Actual House Prices:
     [0.477   0.458   5.00001 2.186   2.78    1.587   1.982   1.575   3.4
      4.466  ]

     Predicted House Prices:
     [0.71912284 1.76401657 2.70965883 2.83892593 2.60465725 2.01175367
      2.64550005 2.16875532 2.74074644 3.91561473]

     Mean Squared Error: 0.5558915986952422
     R2 Score: 0.5757877060324524
```

SAMPLE INPUT

Feature	**Value**
Area	1500 sq.ft
Bedrooms	3
Bathrooms	2
Floors	1
Location	Urban

SAMPLE OUTPUT

House Price Prediction

Predicted House Price: ₹65,00,000

Model Performance:
Mean Squared Error: 2.15
R^2 Score: 0.91

RESULT

The **House Price Prediction Model was successfully implemented using Linear Regression**. The model predicts the approximate price of a house based on its given features with good prediction accuracy.